

Emergency Predictor

Adina Busca

May 2025

Contents

1	Introduction	2
2	Technologies	2
3	Detailed Description	2
3.1	Machine Learning	2
3.2	Emergency Logic	3
3.3	User Input via GUI	4
3.4	User Data Logging	4
3.5	Retraining the Model (Future Work)	4

1 Introduction

Emergency Predictor is an AI system that predicts whether a user is experiencing a medical emergency based on their age and severity of symptoms. It combines:

- A Random Forest ML model
- A Tkinter-based GUI
- A data logger
- Synthetic and real user data

2 Technologies

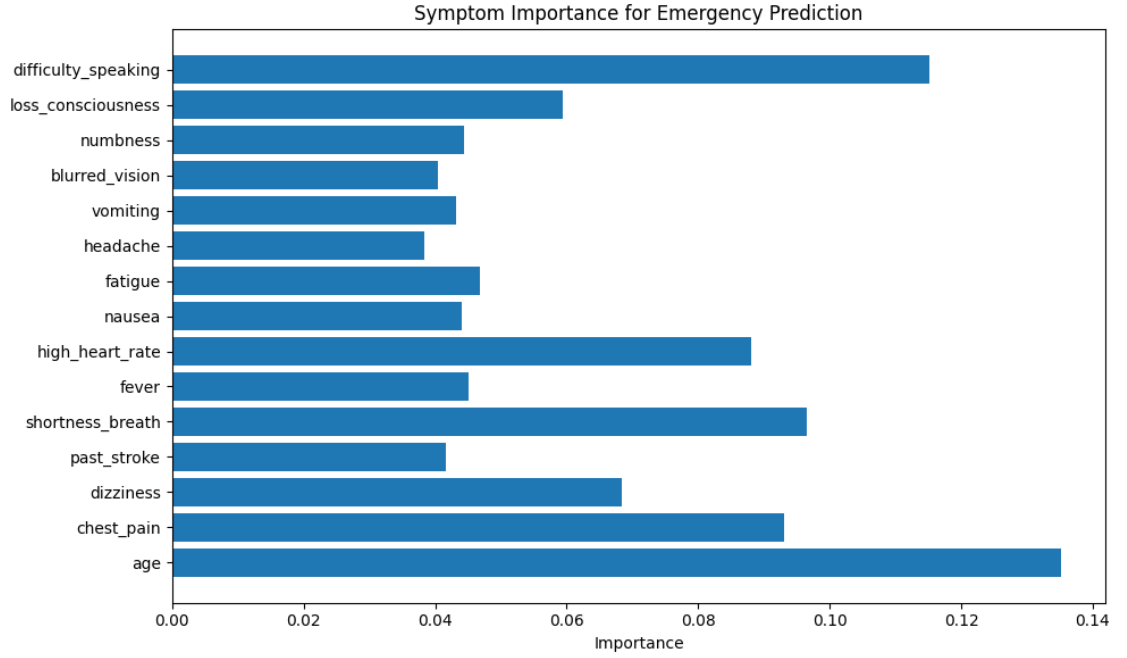
Emergency Predictor is made in Python language. For the making of Emergency Predictor, the following technologies have been used:

- scikit-learn - for Machine Learning
- pandas - for Data Processing
- joblib - for Model persistence
- matplotlib - for Feature Importance chart
- tkinter - for GUI interface
- csv/os - for Data logging and file management

3 Detailed Description

3.1 Machine Learning

The machine learning model was trained on 300 mock patient cases (with random symptoms) using the Random Forest algorithm (100 decision trees). The model was trained on 80% of the available data and evaluated on the remaining 20% for its accuracy. After training, the trained model is saved for future use and feature importance is shown in a bar chart in order to check which symptoms contributed most to emergency prediction.



3.2 Emergency Logic

For this project, I chose one of the most common symptoms (14), of which 5 were classified as critical. This symptoms can hold a severity value from 0(none) to 5(severe).

In order to obtain the emergency (0/1), the following are taken into account:

- score sum of all of the symptoms / 70 (total score)
- score sum of all of the critical symptoms / 25 (critical total score)
- the age risk (higher (1) if the user is older than 50, lower otherwise (0))

Afterwards, the base probability is computed as follows:

- 40% : total score
- 50% : critical total score
- 10% : age risk

The emergency is:

- 1, if the base probability is greater than 0.6, meaning that there is an emergency
- 0, otherwise, meaning that there is no emergency.

3.3 User Input via GUI

The user is asked to give their age and the severity (0-5) of 14 symptoms. After pressing the Predict button, the user is presented with a message that informs them whether their situation is an emergency and the probability it (the confidences score).

3.4 User Data Logging

After each use, the details and results obtained are saved in a file that is meant for future retraining of the model.

3.5 Retraining the Model (Future Work)

Though not yet implemented since the number of user has not reached a significant number, the retraining will be done by combining the mock (the one done with the 300 samples) with the real data obtained from the users.